Software Test Plan (STP)

Perishables Management System

Project	**Perishables Management System (PMS)**
Version	1.0
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Status	Draft — partially implemented; execution results pending

Revision History

Version	Date	Author	Change summary
1.0	28 October 2026	Four-member Software Engineering project team	Initial project-specific draft

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1. Introduction

1.1 Purpose

This plan defines the scope, strategy, environment, responsibilities, and exit criteria for testing PMS. Tests verify that inventory and batch data are managed correctly, stock movements are consistent, expiry/low-stock alerts are accurate, and access controls are enforced.

1.2 Scope

Testing covers authentication/authorization, product CRUD, batch receipt, quantity/date validation, stock reductions and waste, stock calculations, expiry and low-stock alerts, FEFO sorting, reports, audit events, and key non-functional requirements. Real retail POS integration, hardware sensors, external messaging, and production-scale infrastructure are excluded unless implemented.

2. Test Items

- Authentication and role authorization
- Product management module
- Batch and inventory management module

- Stock movement and wastage module
- Expiry and low-stock alert module
- Reporting module
- Audit history
- Application routes, services, and persistence layer

3. Features to be Tested

Requirement / feature	Test focus
PMS-F-001 to PMS-F-002	Valid/invalid login, logout, role restrictions, direct protected-operation access.
PMS-F-003 to PMS-F-004	Create, view, edit, search, filter, and validation of product fields.
PMS-F-005 to PMS-F-007	Batch receipt, date/quantity validation, and stock-total calculations.
PMS-F-008 to PMS-F-009	Stock reduction/waste, insufficient-stock rejection, and authorized correction.
PMS-F-010 to PMS-F-012	Near-expiry, expired, and low-stock classification at boundary values.
PMS-F-013 to PMS-F-014	Waste history and FEFO ordering.
PMS-F-015 to PMS-F-017	Inventory, wastage, near-expiry, and expired-stock report contents.
PMS-F-018 / PMS-SR-005	Audit event content and exclusion of secrets.
PMS-NF-001	Dashboard/search response time on the agreed test dataset.
PMS-NF-002 / PMS-NF-006	Persistence after restart and documented setup on a clean environment.
PMS-NF-003 / PMS-SR-001 to PMS-SR-004	Credential storage, authorization, validation, and safe error responses.

4. Features Not to be Tested

- Physical freshness or quality of fruit/vegetables; the system relies on user-entered data.
- External supplier, POS, barcode scanner, weighing scale, email/SMS, or ERP integrations that are not part of the submitted build.
- Production infrastructure availability and load at real retail-chain scale.
- Third-party database internals; test only the application's use of the selected database.

5. Test Approach / Strategy

5.1 Test Levels

- Unit testing: validate service-level rules such as date validation, stock calculations, insufficient-stock handling, and alert classification.
- Integration testing: validate service/repository/database interactions and transactional stock updates.
- System testing: run end-to-end workflows through the application interface or API.
- Acceptance testing: demonstrate the main user stories against the submitted SRS and verify the working product.

5.2 Test Types

- Functional testing of each implemented requirement.
- Negative and boundary-value testing for quantity, date, permissions, and empty data.
- Regression testing after service, UI, or schema changes.
- Security checks for authentication, authorization, password storage, input validation, and logs.
- Performance checks for dashboard/search using a documented dataset.
- Usability checks for clear navigation, labels, validation feedback, and status indicators.

5.3 Entry Criteria

- A runnable build is available and setup steps are documented.
- Test environment and test database are available.
- The team has agreed the implemented requirement set and test data.
- Critical dependencies are installed and automated tests can be executed.

5.4 Exit Criteria

- All planned high-priority test cases have been executed.
- All critical defects are fixed or explicitly accepted by the team/instructor; no known critical stock-integrity or authorization defect remains.
- Stock totals reconcile for the test workflows.
- Implemented requirements are linked to test cases in the RTM.
- Test results and deviations from SRS/SAD are documented.

6. Test Environment

Area	Plan
Hardware	Team development laptops/desktops; no special hardware required.
Software	The same Python runtime, virtual environment, application framework, and database specified by the project README.
Browser	Current version of a Chromium-based browser; add other browsers if required by the team.
Tools	Project unit-test framework, GitHub Actions CI, browser developer tools, and issue/PR tracking in GitHub.
Test data	Synthetic products and batches with past, current, and future expiry dates; stock quantities covering normal, low, zero, and insufficient-stock cases.
Isolation	Use a test database or resettable fixtures; do not run destructive tests against real store data.

7. Test Schedule

Activity	Target window	Output
Review SRS/SAD and prepare cases	Before / during Sprint 1	Test cases and RTM updates
Unit and integration tests for implemented stories	Sprint 1 (19–23 Oct 2026, per course brief)	Automated test results and defect list
End-to-end regression and alert/report tests	Sprint 2 (26–30 Oct 2026, per course brief)	Execution log and resolved defects
Final acceptance run and evidence capture	By end of Sprint 2	Test summary, screenshots/video evidence, final RTM

Dates above follow the course schedule supplied by the team. Adjust actual execution dates if work or submission dates change.

8. Test Deliverables

- This Software Test Plan.
- Test cases and test data/fixtures.
- Automated unit and integration test scripts.
- Test execution logs and CI results.
- Defect/issue records with reproduction steps.
- Requirements Traceability Matrix (RTM).
- Final Test Summary Report including pass/fail counts, known issues, and deviations.

9. Roles and Responsibilities

Role	Assigned person	Responsibilities
Test coordinator / QA	Deeksha	Maintain plan and RTM; coordinate test execution and report status.
Backend / service developer	Chris	Add service unit tests; fix service and data-integrity defects.
UI / integration developer	Janhavi	Verify UI workflows and integration behaviour; fix UI defects.
CI / documentation / cross-functional tester	Chiranth	Maintain CI checks, run regression tests, collect evidence, and update final documentation.

All four members should review one another's pull requests and run relevant tests before merging. These are suggested responsibilities; replace them with the team's actual assignments.

10. Risks and Mitigation

Risk	Mitigation
Late integration causes failures	Merge small PRs frequently and run CI on every PR.
Test data does not cover expiry boundaries	Include expired, expiring today, expiring within threshold, and outside-threshold batches.
Stock totals drift after multiple operations	Use transactional service logic and reconciliation tests.
Tests depend on local configuration	Document environment setup and keep secrets out of the repository.
Requirements changed during implementation	Record each approved deviation and update SRS, RTM, and test cases.

11. Assumptions and Dependencies

- The project build and dependencies can be installed using the repository instructions.
- Test data is synthetic and can be reset between runs.
- All four team members have access to the shared repository and can review/merge changes according to the team's workflow.
- Any external integration not present in the build is treated as out of scope.

12. Suspension and Resumption Criteria

Suspend a test run if the application cannot start, the test database is unavailable, a blocker prevents more than 30% of planned cases from running, or a test risks corrupting shared data. Resume when the blocking defect is fixed, the environment is restored, and a smoke test passes.

13. Test Case Management and Traceability

Test case ID	Scenario	Expected result	Requirement
TC-AUTH-01	Sign in with valid credentials	User is authenticated and sees permitted dashboard.	PMS-F-001
TC-AUTH-02	Attempt protected action without permission	Request is rejected; no data changes.	PMS-F-002, PMS-SR-002
TC-PROD-01	Create product with valid required fields	Product is saved and visible in search.	PMS-F-003
TC-PROD-02	Submit product with missing name	Validation message shown; no invalid product saved.	PMS-F-003, PMS-F-020
TC-BATCH-01	Receive batch with positive quantity and valid dates	Batch and receipt movement are persisted.	PMS-F-005, PMS-F-007
TC-VAL-01	Submit zero/negative quantity	Request is rejected; stock remains unchanged.	PMS-F-006
TC-VAL-02	Expiry date earlier than received date	Request is rejected with validation feedback.	PMS-F-006
TC-STOCK-01	Reduce stock by an available quantity	Stock decreases by the requested amount and movement is recorded.	PMS-F-008
TC-STOCK-02	Reduce stock by more than available quantity	Operation is rejected; stock remains unchanged.	PMS-F-009
TC-ALERT-01	Batch expires inside configured warning window	Batch appears in near-expiry alerts.	PMS-F-010
TC-ALERT-02	Batch is expired and has remaining quantity	Batch appears in expired-stock view.	PMS-F-011
TC-ALERT-03	Product quantity is below minimum threshold	Product appears in low-stock alerts.	PMS-F-012
TC-FEFO-01	Sort multiple batches of same product by expiry	Earliest expiry batch appears first.	PMS-F-014
TC-REP-01	Generate inventory report for a defined scope	Report totals match stored stock data.	PMS-F-015
TC-REP-02	Generate wastage report for a date range	Only matching waste movements are included.	PMS-F-016
TC-AUDIT-01	Record a stock adjustment	Audit record includes actor/action/time and no secrets.	PMS-F-018, PMS-SR-005
TC-PERF-01	Measure 20+ dashboard/search requests on 5,000 batches	At least 90% complete within 2 seconds.	PMS-NF-001
TC-SEC-01	Inspect stored credential representation and logs	No plaintext password or secret is stored/logged.	PMS-NF-003, PMS-SR-003
TC-RESTART-01	Restart application after committing data	Previously committed records remain available.	PMS-NF-002

14. Test Metrics and Reporting

- Execution progress = executed test cases / planned test cases × 100%.
- Pass rate = passed test cases / executed test cases × 100%.
- Track failed, blocked, and not-run cases separately.
- Track open defects by severity, age, and affected requirement.
- Report requirement coverage as requirements with at least one test / implemented requirements.
- Publish daily or per-PR status during the sprint and a final test summary before the demo.